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Canadian Association of Radiologists
L'Association canadienne des radiologistes

POSITION STATEMENT (Updated 2026)

Establishing a National, Sustainable and Secure Canadian Helium Supply

October 2026

EXECUTIVE SUMMARY

- Canada's healthcare and research systems depend on a stable and secure supply of liquid helium, yet this supply remains fragile and heavily dependent on external factors.
- Despite rapid growth in domestic helium production—reaching approximately 5% of global supply—Canada lacks large-scale purification and liquefaction capacity, forcing reliance on a costly and insecure "export and re-import" model.
- Global supply volatility, geopolitical risks, and protectionist policies continue to threaten more than 610 MRI and NMR systems across Canada.
- Investments in helium recovery and recycling systems can significantly reduce demand pressures and strengthen system resilience in the near term.
- Strategic federal action can establish a sovereign Canadian helium supply chain, protecting critical infrastructure and supporting national healthcare and research priorities.

THE ISSUE

Over the past decade, disruptions in the global helium market have become more frequent and severe.[1-3] Supply shortages have been driven by geopolitical tensions, infrastructure outages, and the highly concentrated nature of global production. These dynamics have caused both supply instability and sharp price fluctuations, directly affecting Canadian helium users.[4,5]

While upstream helium gas extraction, particularly in Saskatchewan, has expanded rapidly, the country lacks the midstream infrastructure required to purify and process helium into its usable pure liquid form. Almost all Canadian helium gas must therefore be exported for liquefaction before being re-imported for domestic use. This "buy-back" model not only introduces avoidable costs and supply risks, but entrenches

a structural vulnerability, placing critical healthcare and research infrastructure at the mercy of global markets. Canada remains vulnerable within this system.

At the same time, this structural weakness presents a clear strategic opportunity. By closing this gap, Canada can align its helium resources with its broader critical minerals strategy and build a resilient, self-sufficient supply chain.

RECOMMENDATIONS

To address these challenges, the Canadian Helium Users Group (CHUG) calls for a coordinated federal response:

1. Supply-Side Investment (Sovereign Capability)
The Government of Canada should work with provincial governments, Indigenous partners, and industry to accelerate the development of large-scale helium purification and liquefaction capacity within Canada. This investment would complete the domestic supply chain and eliminate reliance on foreign processing.

2. Demand-Side Mitigation (Recovery and Recycling)
The federal government should invest $40 million over four years, administered through Innovation, Science and Economic Development Canada, to support the installation and optimization of helium recovery and recycling systems across healthcare and research institutions, where possible.

WHY HELIUM MATTERS

Helium is a non-renewable critical mineral with no substitute in key applications. Its most essential use in Canada is for the operation of superconducting magnets found in Magnetic Resonance Imaging (MRI) scanners and Nuclear Magnetic Resonance (NMR) spectrometers. Without a continuous supply of liquid helium, these systems cannot function.

A disruption in supply does not simply pause operations—it will lead to a "quench," a catastrophic failure in which magnets lose superconductivity and may be permanently damaged. The cost of replacing a single system ranges from $1.2 million to $10 million, emphasizing the magnitude of the risk.

Beyond healthcare and research, helium is also essential in advanced manufacturing sectors such as semiconductors, aerospace, deep sea diving, nuclear power plants and quantum technologies, further amplifying its strategic importance.

RISK TO HEALTHCARE AND RESEARCH

The implications of helium supply instability are particularly acute in healthcare. Medical radiation technologists (MRTs) provide the essential link between compassionate care and the sophisticated medical imaging and therapeutic technologies that underpin modern healthcare. In total, more than 2.2 million examinations are performed annually in Canada.[6,7] Even short disruptions can have significant

consequences: a two-week shutdown of a single MRI unit can result in hundreds of cancelled scans, delaying diagnoses and treatment for patients.

As provinces continue to invest in expanding MRI capacity, demand for liquid helium will only increase. Without a reliable supply, new systems cannot be commissioned, directly limiting improvements in patient access and care.

The research sector faces equally serious risks. NMR spectroscopy plays a foundational role in fields ranging from drug discovery and life sciences to clean energy and advanced materials. It supports a large and highly skilled research workforce, with CHUG survey data indicating that more than 5,000 highly qualified personnel depend on these systems.

Despite some progress in mitigation, supply challenges persist. In a recent survey or NMR research laboratories in Canada, nearly one third of respondents reported receiving unscheduled price increases and surcharges due to helium supply disruptions. Some of the respondents are already facing helium allocation, while the broader impact of the ongoing crisis in the Middle East and the war between Russia and Ukraine has yet to be felt. These trends threaten both Canada's research capacity and the return on significant public investment in scientific infrastructure. Collectively these research NMR centres support the activities of nearly 4,000 students, post-docs and professors across Canada.

GLOBAL MARKET VULNERABILITY

Canada's exposure is compounded by the structure of the global helium market. Production is highly concentrated, with a significant share originating in the United States and the remainder in regions subject to geopolitical instability. This concentration makes supply chains highly sensitive to political and economic disruption.

Moreover, MRI and NMR systems require very pure helium gas and liquid helium specifically, leaving users with little flexibility in the face of supply constraints. Unlike many industrial applications, there are no viable substitutes, making these sectors extremely vulnerable to shortages.

CANADA'S STRATEGIC OPPORTUNITY

Canada is uniquely positioned to address these vulnerabilities. Saskatchewan's helium gas production has increased more than fourteenfold since 2020, with national output now estimated at approximately 5% of global supply. However, this growth has not translated into domestic supply security due to the absence of large-scale processing infrastructure in Canada.

The central challenge is therefore not resource availability, but infrastructure. Without helium purification and liquefaction capacity, Canada remains dependent on foreign processing and exposed to external market forces.

Year	**Saskatchewan Production (m^3) (Reference 8)**
2020	352,000
2021	1,333,000
2022	2,378,000
2023	3,747,000
2024	4,997,000
2025	**4,956,000 (est.)**

Strategic federal investment, aligned with existing provincial initiatives, can close this gap. Saskatchewan has already implemented targeted incentive programs, and several projects are considered "shovel-ready." By partnering with provinces, industry, and Indigenous groups, the federal government can catalyze the development of a complete, domestic helium value chain.

Such a transformation would secure supply for critical sectors, create economic value within Canada, and reinforce the objectives of the Canadian Critical Minerals Strategy.

REDUCING DEMAND PRESSURE: RECOVERY AND RECYCLING

While supply-side solutions are essential, immediate long-term relief can be achieved through conservation measures. Helium recovery systems offer a proven and highly effective way to reduce consumption. By capturing boil-off gas and re-liquefying it for reuse, these systems can recover up to 95% of helium and reduce overall demand.

Facilities that have implemented recovery systems consistently report reduced exposure to supply disruptions, allocation limits, and price increases. However, adoption across Canada remains uneven, leaving many institutions vulnerable.

Beyond cost savings, these systems provide critical asset protection. By maintaining stable helium levels, they reduce the risk of quench events and the associated financial and operational consequences, which can include repair costs in the hundreds of thousands of dollars and extended downtime.

A coordinated national approach would maximize these benefits. CHUG proposes a hub-and-spoke model in which larger institutions serve as regional liquefaction hubs, supporting smaller facilities that collect and compress helium locally. This model improves efficiency, reduces costs, and enhances resilience across the system.

The financial case for investment is strong, with typical return on investment ranging from two to seven years, often faster in the context of rising helium prices.

ENVIRONMENTAL AND ECONOMIC BENEFITS

Investment in helium infrastructure also delivers important environmental and economic benefits. Recovery systems support a circular economy for a non-renewable resource, reducing waste and limiting emissions associated with global transport. At the same time, Saskatchewan's helium resources are often not tied to hydrocarbon production, offering a comparatively low-emissions supply source.

Developing a domestic supply chain would retain economic value within Canada, strengthen industrial capacity, and support growth in advanced technology sectors.

ABOUT US

The **Canadian Helium Users Group (CHUG)** is a national consortium of researchers, facility managers, and technical experts responsible for the operation and stewardship of MRI and NMR infrastructures across Canada. The group plays a central role in ensuring the reliability and sustainability of systems that underpin both healthcare delivery and scientific research. (https://nmrx.ca/en/chug.html)

The **Canadian Association of Radiologists (CAR)** represents over 3,000 radiologists across Canada. As the national voice for radiology, the CAR is committed to championing safe, high-quality, and timely patient care. Recognizing that stable medical imaging infrastructure is vital to the healthcare system, the CAR actively collaborates with CHUG to advocate for reliable liquid helium access, ensuring that life-saving MRI technologies remain operational and accessible for Canadian patients.

The **Canadian Association of Medical Radiation Technologists (CAMRT)** is the national professional association and certifying body for medical radiation technologists (MRTs) across Canada. CAMRT is a leading advocate and authoritative voice on the critical issues that affect the more than 22,000 MRTs working in medical imaging (radiologic technology, nuclear medicine, magnetic resonance) and radiation therapy.

CONCLUSION

Canada faces both a critical vulnerability and a significant opportunity. The current dependence on foreign helium processing exposes essential infrastructure to unacceptable risk, while domestic production growth provides the foundation for a secure, stable self-sufficient system.

By addressing both supply and demand through targeted federal action, Canada can eliminate this vulnerability, protect its healthcare and research capacity, and establish itself as a leader in a strategically important sector, while saving a critical natural and non-renewable resource.

References

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7. PET-MRI added approximately 1,200 exams among sites with available data, representing a national average of **0.03 exams per 1,000 population.** **(https://canjhealthtechnol.ca/index.php/cjht/article/view/HC0024-04/2068)**

8. **Source:** Petrinex. (2026). *Saskatchewan Public Data: Conventional Volumetric Data Download (2020–2025)* [Data set]. Retrieved April 23, 2026, from https://www.petrinex.ca/public-data/spd/